

Lecture 18

3/2/26
JRA

HK Finish std deblurring
Move to spectrum of special techniques
MT $\approx$ 2 wks

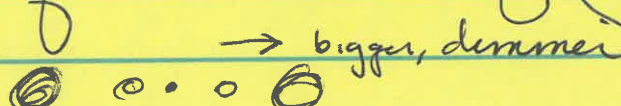
HW - pinhole confocal

LS pinhole in int
SD illum lots of pinholes at one time Fast
less precise

Computational method Comp Deblurring/Deconv

Building up a 3D image from lesser info
is pervasive

Want to know ^{micro's} med 3D distribution of stuff
in object

Microscope Get 3D optical sections
Distorted by diffraction
Get image of Point source of light


Imaging theory recap

$$I = O \otimes \text{PSF}$$

3 things are related know 2
 $\uparrow$ Convolution (an integral)

Know I & PSF

Solve for O

If $\otimes$ were simple mult could solve for O by dividing

Reality: ^{more complicated} Iteration

Start w/ "intelligent guess" for O
 Best guess is $I = O$

Start w/ guess

Computer convolves guess w/ PSF

If guess is good

$$G \otimes \text{PSF} = I$$

Compare convolution w/ image & revise guess

Ex 50 / 100 raise by ratio

Iterate ≈ 20 times get

Guess too high

$$G \times \text{PSF} = 200 \quad I = 100$$

Lower G by $100/200$

~ 20 Iterations

Reconstruct object

Software approach complements HW approach
Can get 0

Often done in conjunction w/ confocal

Detour * Polarization microscopy

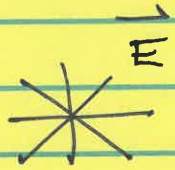
Look at one more very powerful method
of looking at unstained samples

Light is composed of E & B fields

Natural light is coming from many atoms.
Each atom light is polarized but sum is
unpolarized

Natural light is "randomly" polarized

Each atom will generate light w/ a given E direction (electric field)



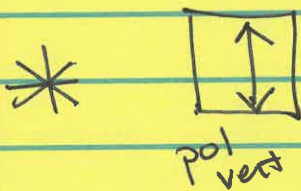
lots of atoms
Randomly polarized

DEMO

One polarizer -
Two polarizers - crossed
Dark



polarizer lets thru vertical light $\vec{E}$



pol
vert



pol
hor

Detector
No Light

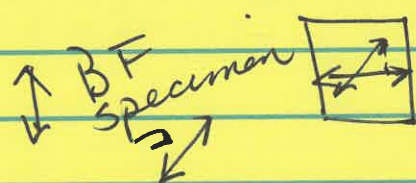
Crossed
polarizers
because
transmission
axes are
crossed

Microscope can have two crossed polarizers
 $\Rightarrow$ dark background

Polarization microscope

No specimen $\Rightarrow$ no signal
dark background

Birefringent samples rotate light polarization



some horizontal light
will get thru

Specimen that is birefringent will create a signal

2) BF specimen will ^{rotate light pol $\hat{e}$} create a ^{bright} signal ^{due to light} against a dark background

Things like microtubules are different $\parallel \hat{e} \perp$ to long axis. These work well w/ pol light

1950's Movie Microscopy

No staining

HW

Rules of the game w/ polarizers

- (1) Unpolarized light hits a polarizer
Output intensity is down by a factor of two
 $I_{out} = I_{in} / 2$

∴ light is polarized along transmission axis

- (2) Polarized light hits a polarizer

Malus's Law



pol along trans axis of second pol

$$I_{out} = I_{in} \cos^2 \theta$$

Polarizer w/
cocked transmission axis

Ex



$\theta = 90^\circ$

Crossed pol = 90°

ACTIVITY 7

Polarizers

	(a)	A	$I_0 / 2$	$\uparrow$ vert
		B	$I_0 / 8$	$\swarrow 60^\circ$
$I_0 / 8 \cos^2 30^\circ$	30°	C	$3I_0 / 32 = 0.094 I_0$	

this is the signal caused by a specimen that rotates light

(b)	A	$I_0 / 2$	crossed pol No signal
	C	0	

1 Vert
2 60°
3 Hor

Reviewed answers

Sample (a) is like having a polarizing specimen

(B) 90° difference

DEMO 2 Add X polarizer

© 2026 Beth A. Scalettar & James R. Abney